EXPERIMENT :- Study of LAN Topology Network devices and address configuration.

OBJECTIVE :-

1.) To study the LAN topology used in college campus.
2.) To identify network device & understand their function.
3.) To analyze Wired & Wired Wireless communication method used in the network.
4.) To measure speed of network for Wired & Wireless connections.
5.) To obtained mac, IP address & subnet mask.

→ Problem statement - 1

- Draw & identify LAN topology of college campus & clearly label all components such as router, switches & etc.

→ Soln :- The college campus network observed this study consists of:

* Layer-3 core switches in the server Room.
* managed by POE + switches
* Wireless Access Point (AP3)
* Ethernet part is classroom & Laboratories.

→ Network Topology.

Hybrid Hierarchical Topology which combines:

1.) Tree Topology
2.) Extended star Topology
3.) Ring Topology (server room)

→ The network hierarchical layer.

(1) Core layer
(2.) Distribution layer
(3.) Access layer

→ Core Layer

* Core layer contains server Room and server room have Firewall, Router, 2-L3+ switches, Path Rack, server, inverter etc.

→ Distribution Layer

* Distribution Layer contains 4 switches
* The 4 switches are →

(1) Main Build → R7 (Room no- 213, 2nd floor).
(2.) Main Build → R1 (Room no. 301, 3rd floor).
(3.) Main Build → R6 (Room no 13, 1st floor)
(4.) Main Build → R2 (Room no. 201, 2nd floor).

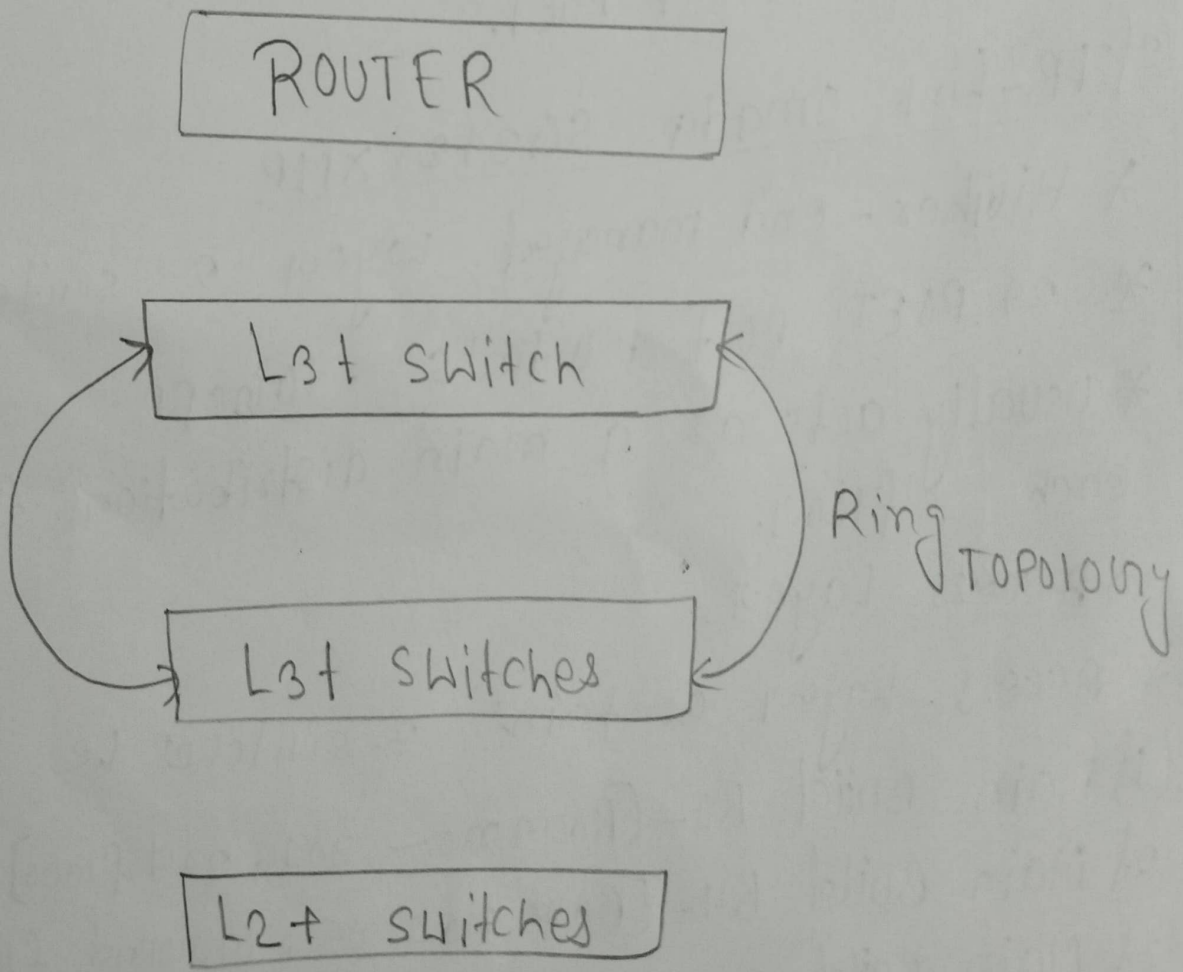
* This 4 switches contains →

① 18 port Gigabit easy smart switch with 16- Port [TL-SG1218MPE]

* Every switches act as access layer which helps to provide network connecting to all class room, Lab, Faculty Room etc.

* Access Layer directly connect:
(1) Ethernet port
(2) Wireless AP
(3) Lab system
(4) Outdoor Wireless AP (ex- OIS)

→ SERVER ROOM ARCHITECHER/TOPOLOGY



② SG3428XMP → 24 Port Gigabit & 4-Port 10GESFP+, L2+ managed switches with 24 Port POE+.

→ Now Let's discuss these 2 above components →

① TP-Link TL-SG1218MPE

* Likely used for: CCTV cammera, Wifi access point, etc.
* 18 Port Gigabit "Easy smart" switch.
* 16 Port support POET.

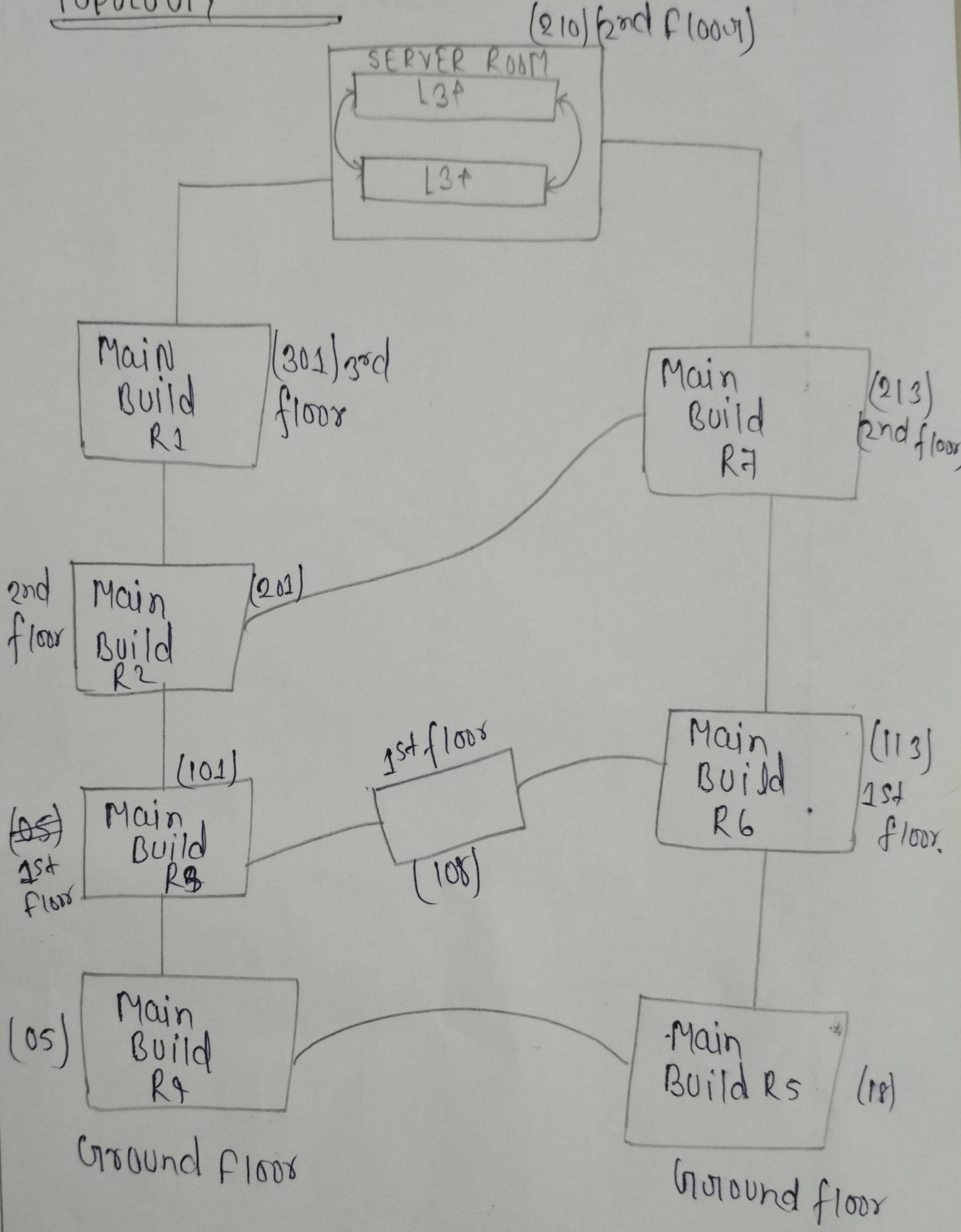
② TP-Link Omada SG3428XMP

* Higher-end managed layer 2+ switch
* 24 POET Port, Unlinks, 10hsFP
* Usually acts as a main distribution for each floor.

→ Access Layer

* Access Layer contains 3-switches i.e
(1) Main Build R1- [Room no- 301, 3rd floor].
(2) Main Build R4- [Room no- 05, Ground floor]
(3) Main Build R5- [Room no- 18- Ground Floor]

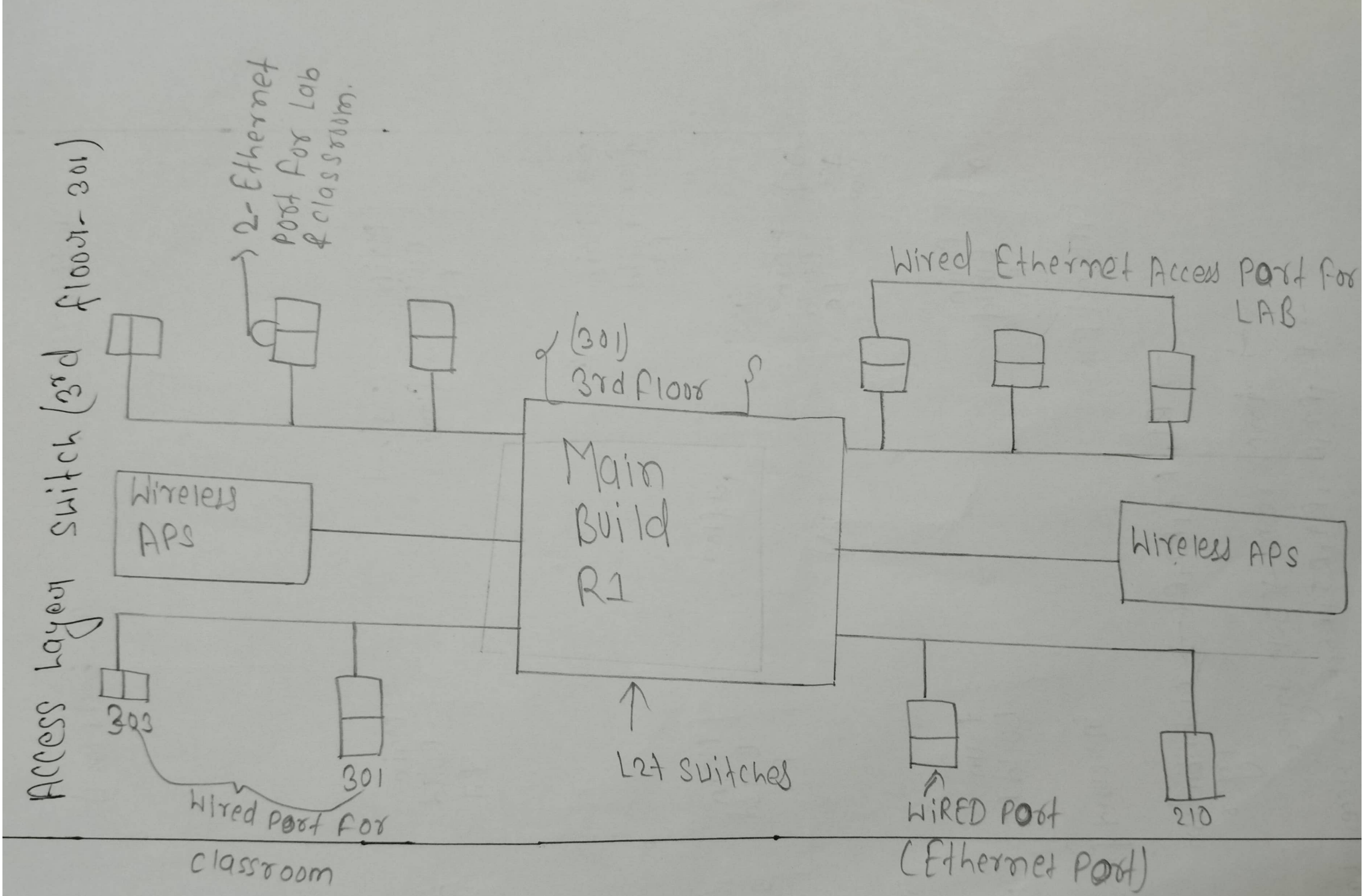
TOPOLOGY



→ R1, R7, R2, R6 → Distribution layer

→ server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access Layer


Access Layer Switch (3rd floor - 301)
2- Ethernet port for 906 classroom
(301)
3rd floor
Main Buil R1
Wireless APS
Wired Ethernet Access Port for LAB
Wireless APS
303
301
Wired Port For classroom
L2+ Switches
WIRED Port (Ethernet Port)
210

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hp> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d309:59fc:2062:ffee%13
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi 2:

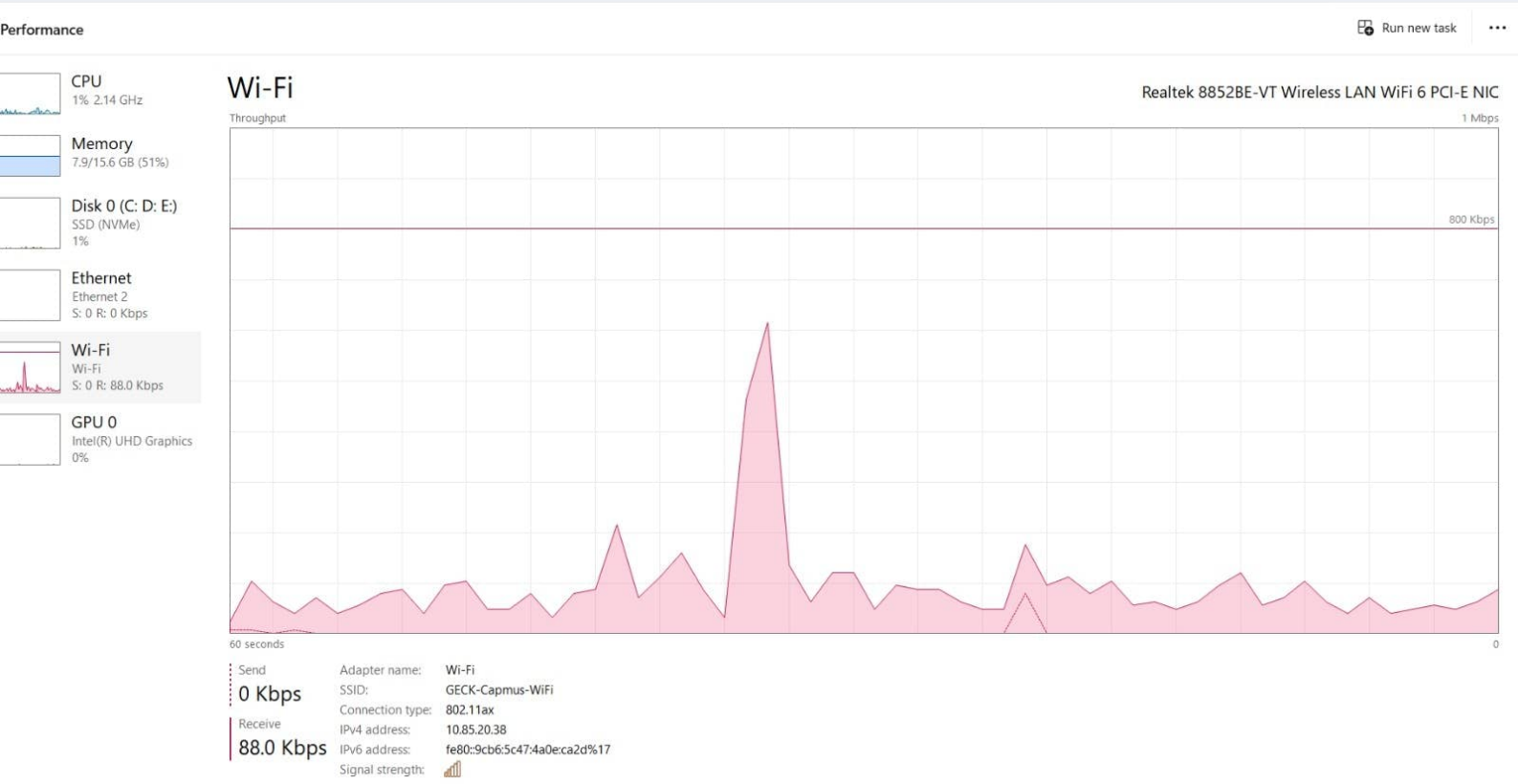
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 5:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9cb6:5c47:4a0e:ca2d%17
   IPv4 Address. . . . . . . . . . . : 10.85.20.38
   Subnet Mask . . . . . . . . . . . : 255.255.248.0
   Default Gateway . . . . . . . . . : 10.85.16.1
PS C:\Users\hp> S
```


Performance
Run new task
CPU
1% 2.14 GHz
Memory
7.9/15.6 GB (51%)
Disk 0 (C: D: E:)
SSD (NVMe)
1%
Ethernet
Ethernet 2
S: 0 R: 0 Kbps
Wi-Fi
Wi-Fi
S: 0 R: 88.0 Kbps
GPU 0
Intel(R) UHD Graphics
0%
Wi-Fi
Realtek 8852BE-VT Wireless LAN WiFi 6 PCI-E NIC
Throughput
1 Mbps
800 Kbps
60 seconds
0
Send
0 Kbps
Receive
88.0 Kbps
Adapter name: Wi-Fi
SSID: GECK-Capmus-WiFi
Connection type: 802.11ax
IPv4 address: 10.85.20.38
IPv6 address: fe80::9cb6:5c47:4a0e:ca2d%17
Signal strength:

```
PS C:\Users\hp> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2409:40e4:205a:90f5:651d:429a:9417:302
   Temporary IPv6 Address. . . . . . : 2409:40e4:205a:90f5:b419:51fd:3072:24c4
   Link-local IPv6 Address . . . . . : fe80::da50:1978:6733:c4dc%46
   IPv4 Address. . . . . . . . . . . : 10.160.177.67
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::887:aff:fe25:9fb1%46
                                       10.160.177.122

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d309:59fc:2062:ffee%13
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 5:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
PS C:\Users\hp> ^SSśś
```

→ Difference b/w Wired and Wireless speed.?

Soln:→

Wired Network	Wireless Network
* Faster communication	* slower communication
* stable connection	* affected by distance/Interference
* Lower latency	* Higher latency
* dedicated physical medium.	* shared radio medium

→ Ethenet provides direct physical communication using network cables resulting in higher speed & stable performance.

→ wireless connections uses radio waves which are affected by interference distance, obstacles & multiple connected users.

Q. Role of subnet mask in network?

Soln:→ A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another devices belongs to the same network or different network.

* It is Important for :-
  (1) Network segment
  (2) Efficient routing
  (3) Communication between ~~hardware~~ devices.

→ OBSERVATION:-

During the study of the college network infrastructure:-

(i) A centralized server room was used as network backbone.

(ii) TWO L3 switches were connected in redundant cyclic structure.

(iii) Managed TP-Link POET switches wher used in network cabinets.

→ Advantage of Existing network?

* Scalable architecture
* centrallized management
* supports POET devices
* Easy Explanation of floors & classroom.
* support both wireless & wired
* Redundant link for failovers.

CONCLUSION:-

The observed campus LAN follow a Hybrid hierarchical topology consist of tree Extended star, Ring. the network wes manged POE switches, wireless access point Ethernet Port & centralized server infrastructure to provide efficient wired & wireless communication accross the campus.